# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE NAME: Artificial Intelligence** **COURSE CODE: CSA1741**

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**COURSE OUTCOME COVERED**

***CO1:*** Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

*Assessment Tool Weightage: 10%*

**QUESTIONS:2** **Total Marks:20**

## Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

# Question 1: Types of Intelligent Agents in Smart Home Automation

## Introduction

An intelligent agent is a system that perceives its environment through sensors and performs actions using actuators to achieve a particular objective. In a smart home, it controls lighting, temperature, and security based on user behaviour and environmental conditions.

## 1. Simple Reflex Agent

- Acts according to the current input using condition-action rules.
- Does not remember past experiences.
- Example: If motion is detected, the light turns ON. If no motion is detected, the light turns OFF.
- Advantage: Fast and easy to implement.
- Disadvantage: Cannot learn user preferences or use past information.

## 2. Model-Based Agent

- Maintains an internal state of the environment using current and previous information.
- Example: Remembers that a person is inside a room even if the motion sensor temporarily stops detecting movement.
- Advantage: Handles partially observable environments.
- Disadvantage: Requires additional memory and computation.

## 3. Goal-Based Agent

- Selects actions to achieve a specific goal.
- Example: Adjusts the air conditioner to maintain the user's desired temperature.
- Advantage: Can plan actions to achieve a target.
- Disadvantage: Planning may require more time and computation.

## 4. Utility-Based Agent

- Selects actions based on a utility function that compares different outcomes.
- Example: Balances comfort, electricity consumption, and home security when controlling appliances.
- Advantage: Helps choose the most beneficial action among alternatives.
- Disadvantage: Designing the utility function can be complex.

## Comparison Table

| Agent | Function | Example |
|---|---|---|
| Simple reflex | Reacts to current input | Motion-controlled lighting |
| Model-based | Maintains internal state | Tracks room occupancy |
| Goal-based | Achieves a target | Maintains desired temperature |
| Utility-based | Balances multiple benefits | Saves energy while maintaining comfort |

### Most Effective Approach

A hybrid model-based and utility-based agent is the most effective approach for smart home automation.

- The model-based agent remembers user preferences and room occupancy.
- The utility-based agent balances comfort, energy efficiency, and security.
- Goal-based planning can be added to achieve specific temperature and lighting targets.
- Simple reflex rules can handle urgent actions, such as activating an alarm when a sensor detects an immediate threat.

### Conclusion

A hybrid intelligent agent provides quick responses, learns user behaviour, reduces energy consumption, and improves home security. Therefore, it is the best choice for smart home automation.

## Question 2: Robot Navigation in an Unknown Maze Using UCS and IDS

### Introduction

A robot navigating an unknown maze must find a path to the exit without prior knowledge of the layout. Uninformed search algorithms explore possible paths without using heuristic information about the distance to the goal. Two important algorithms are Uniform Cost Search (UCS) and Iterative Deepening Search (IDS).

### 1. Uniform Cost Search (UCS)

UCS expands the node with the lowest total path cost, represented by \(g(n)\). It uses a priority queue to select the next node.

Steps:

1. Start from the initial position.
2. Add reachable positions to a priority queue.
3. Select the position with the lowest accumulated cost.
4. Explore its unvisited neighbours and update their costs.
5. Continue until the exit is reached.

Advantages:

- Finds the minimum-cost path when step costs are non-negative.
- Suitable when different paths have different costs.
- Complete under the usual finite-branching and positive minimum step-cost assumptions.

Disadvantages:

- Requires large amounts of memory.
- May explore many unnecessary paths.
- Can be slow in large mazes.

## 2. Iterative Deepening Search (IDS)

IDS repeatedly performs depth-limited depth-first search, increasing the depth limit until the exit is found.

Steps:

1. Set the depth limit to zero.
2. Search all positions within that limit.
3. If the exit is not found, increase the limit by one.
4. Repeat the search with the new limit.
5. Stop when the exit is found.

Advantages:

- Requires less memory than UCS.
- Complete when the branching factor is finite.
- Useful when the solution depth is unknown.

Disadvantages:

- Repeats exploration of shallow nodes.
- Does not guarantee the minimum-cost path when step costs differ.
- Can be inefficient for some large search spaces.

## Time and Space Complexity

Let \(b\) be the branching factor, \(d\) the depth of the shallowest solution, \(C^*\) the optimal solution cost, and \(\epsilon\) the minimum positive step cost.

| **Algorithm** | **Time complexity** | **Space complexity** |
|---|---|---|
| UCS | \(O(b^{1+\lfloor C^*/\epsilon\rfloor})\) | \(O(b^{1+\lfloor C^*/\epsilon\rfloor})\) |
| IDS | \(O(b^d)\) | \(O(bd)\) |

UCS generally needs more memory, while IDS is more memory-efficient.

## Relationship to Intelligent Agents

- Simple reflex agent: Responds only to current sensor readings. It may repeatedly visit the same locations.
- Model-based agent: Stores information about discovered walls and visited positions, helping avoid repeated exploration.
- Goal-based agent: Chooses actions that lead toward the exit.
- Utility-based agent: Considers path cost, time, battery consumption, and risk when selecting a route.

## Most Effective Combination

A model-based, goal-based agent using UCS is suitable when finding the minimum-cost route is important.

- The model-based agent remembers the discovered maze layout.
- The goal-based agent focuses on reaching the exit.
- UCS selects the minimum-cost path under the appropriate search conditions.

If memory is limited and all moves have equal cost, IDS combined with a model-based agent may be preferable.

## Conclusion

UCS is best when path cost and optimality are important, whereas IDS is useful when memory is limited and the solution depth is unknown. Combining a suitable search strategy with model-based and goal-based decision-making improves robot navigation.

***Code of Conduct:***

*I Madhan R\192412370 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***
Madhan R

**RUBRICS FOR CALCULATION:**

| Criterion | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) | Max Marks |
|---|---|---|---|---|---|
| **Criteria** | **Level 4 (Exemplary)** | **Level 3 (Proficient)** | **Level 2 (Developing)** | **Level 1 (Beginning)** | **Max Marks** |
| Understanding of Concepts | Demonstrates thorough, accurate understanding of concepts | Good understanding with minor gaps | Basic understanding with some misconceptions | Poor understanding; major misconceptions | 5 |
| Explanation | Detailed, well-explained answers with relevant examples | Clear explanation with adequate details | Limited explanation; lacks depth | Poor or unclear explanation | 5 |

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|---|---|---|---|---|---|
| Application | Effectively applies concepts with strong analytical ability | Applies concepts with minor errors | Limited application with weak analysis | No application or incorrect analysis | 4 |
| Organization | Well-structured answers with logical flow and coherence | Organized with minor issues in flow | Basic structure, lacks clarity | Poor organization, incoherent answers | 3 |
| Accuracy | Completely accurate and covers all required points | Mostly accurate with minor omissions | Partially correct with missing points | Incorrect answers with major gaps | 2 |
| Clarity | Clear, neat, professional presentation | Understandable with minor issues | Limited clarity, readability issues | Poorly written, difficult to understand | 1 |